Yahia DJEMMADA,

Birth-year: 1991.
Citizenship: Algerian.
Marital status: Married.
Work: RECITS Lab., USTHB & National School of Mathematics, Algiers.
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LinkedIn



Education

February 2021 — **Doctor of Philosophy in Mathematics**
Option: *Discrete Mathematics and Operations Research*
University of Science and Technology Houari Boumediene, Algiers, Algeria
Thesis title: *An arithmetic and combinatorial study of the geometric polynomials*
Thesis language: *English*

June 2015 — **Master of Science in Mathematics**
Option: *Operations Research (Methods and Models for Engineering and Research)*
University of Science and Technology Houari Boumediene, Algiers, Algeria
Thesis title: *"Rotating Disk," A symmetric cryptosystem based on the factorization problem*
Thesis language: *French*

July 2013 — **Bachelor in Mathematics**
Option: *Operations Research*
University of Science and Technology Houari Boumediene, Algiers, Algeria
Language of study: *French*

Teaching Experience

Associate Professor (Maître de Conférences A), (January 2025–Present)

- "Discrete Mathematics 1" at The National School of Mathematics, Algiers, Algeria, (September 2025–January 2026).
- "Algebra 1" at The National School of Mathematics, Algiers, Algeria, (September 2025–January 2026).
- "Discrete Mathematics 2" at The National School of Mathematics, Algiers, Algeria, (January 2025–July 2025).
- "Mathematical Computing Tools" at The National School of Mathematics, Algiers, Algeria, (January 2025–July 2025).

Teaching Experience (continued)

Assistant Professor (Maître de Conférences B), (April 2022–January 2025)

- "Discrete Mathematics 1" at The National School of Mathematics, Algiers, Algeria, (September 2024–January 2025).
- "Discrete Mathematics 2" at The National School of Mathematics, Algiers, Algeria, (January 2024–July 2024).
- "Algebra 2" at The National School of Mathematics, Algiers, Algeria, (January 2024–July 2024).
- "Discrete Mathematics 1" at The National School of Mathematics, Algiers, Algeria, (September 2023–January 2024).
- "Algebra 1" at The National School of Mathematics, Algiers, Algeria, (September 2023–January 2024).
- "Mathematical Computing Tools" at The National School of Mathematics, Algiers, Algeria, (January 2023–July 2023).
- "Algorithms and Data Structures 2" at The National School of Mathematics, Algiers, Algeria, (January 2023–July 2023).
- "Algorithms and Data Structures 1" at The National School of Mathematics, Algiers, Algeria, (September 2022–January 2023).
- "Algebra 1" at The National School of Mathematics, Algiers, Algeria, (September 2022–January 2023).
- "Discrete Mathematics 2" at The National School of Mathematics, Algiers, Algeria, (February 2022–July 2022).
- "Python Programming" at The National School of Mathematics, Algiers, Algeria, (February 2022–July 2022).

Assistant Professor (Maître Assistant B), (April 2021–April 2022)

- "History of Mathematics" for 1st-year students of "Mathematics and Informatics" at University of Science and Technology Houari Boumediene, Algiers, Algeria, (April 2021–October 2021).
- "Numerical Analysis" at the Faculty of Civil Engineering, University of Science and Technology Houari Boumediene, Algiers, Algeria, (April 2021–October 2021).

Adjunct Lecturer / Teaching Assistant (Vacataire)

- "Linear Algebra" at The National School of Artificial Intelligence, Algiers, Algeria, (January 2024–July 2024).
- "Algebra 1" at The National School of Mathematics, Algiers, Algeria, (October 2021–February 2022).

Teaching Experience (continued)

- "Mathematics, Statistics, and Informatics" at the Faculty of Nature and Life Sciences, University of Algiers 1, Algeria, (September 2021–February 2022).
- "Mathematics, Statistics, and Informatics" at the Faculty of Nature and Life Sciences, University of Algiers 1, Algeria, (September 2017–February 2018).
- "Mathematics, Statistics, and Informatics" at the Faculty of Nature and Life Sciences, University of Algiers 1, Algeria, (September 2016–February 2017).

Scientific researchs

Published papers

2026 **Djemmada, Y.**, Mehdaoui, A., Németh, L., & Szalay, L. "The Fibonacci-Fubini and Lucas-Fubini numbers." *The Fibonacci Quarterly*.

2024 **Djemmada, Y.**, Mehdaoui, A., Németh, L. & Szalay, L. "An identity for two sequences and its combinatorial interpretation." *Annales Mathematicae et Informaticae*, 60, 37-43.

2023 Belbachir, H. & **Djemmada, Y.**. "The (l, r)-Stirling numbers: a combinatorial approach." *Filomat*, 37(8), 2587-2598.

Belbachir, H. **Djemmada, Y.** & Németh, L. "The deranged Bell numbers." *Mathematica Slovaca*, 73(4), 849-860.

2022 Belbachir, H., **Djemmada, Y.** & Hadj-Brahim, S. "Unified Bernoulli-Euler polynomials of Apostol type". *Indian Journal of Pure and Applied Mathematics*, 1-8.

2021 Belbachir, H. & **Djemmada, Y.** "On central Fubini-like numbers and polynomials". *Miskolc Mathematical Notes*, 22(1), 77-90.

2020 Belbachir, H. & **Djemmada, Y.** "Generalized geometric polynomials via Steffensen's generalized factorials and Tanny's operators". *Indian Journal of Pure and Applied Mathematics*, 51(4), 1713-1727.

2019 Belbachir, H. & **Djemmada, Y.** "Congruence for geometric polynomials". Les Annales RECITS, Vol 06, 37-44.

Accepted papers

2026 Belbachir, H. & **Djemmada, Y.**. "On Meštrović's Generalization of the Chu–Vandermonde Identity." *Filomat*, (accepted).

2025 **Djemmada, Y.**, Kargın, L., & Can, M. "Partial Deranged Bell Numbers and Their Combinatorial Properties." mediterannian journal of mathematics (accepted).

Preprints

Conferences & scientific activities

December 2025 Head of Organization Committee at **The National Conference on Number Theory and It's Interactions (NCNTI25)**, National Higher School of Mathematics, Algiers, Algeria.

July 2024 Leader of the Benjamin exam at the **First Algerian Olympiad of Mathematics (ALMO24)**, organized by the Ministry of National Education, Algiers, Algeria.

Conferences & scientific activities (continued)

June-July 2024 — Trainer and Corrector at the **First Algerian Olympiad of Mathematics (ALMO24)**, organized by the Ministry of National Education, Algiers, Algeria.

January 2024 — Organizer at **R Programming training days**, University of Science and Technology Houari Boumediene, Algiers, Algeria.

November 2023 — Organizer & Tutor at **Python and Sagemath training days**, University of Science and Technology Houari Boumediene, Algiers, Algeria.

Speaker at the **Scientific Days of Laboratory**, University of Science and Technology Houari Boumediene, Algiers, Algeria. (Title: On the (l, r)-partitons and (l, r)-permutations numbers).

May 2023 — Organizer & Judge at **COMPACT-2023** event, National Higher School of Mathematics, Algiers, Algeria.

August 2021 — Speaker at the **10th International Eurasian Conference on Mathematical Sciences and Applications**, Sakarya University, Sakarya, Turkey. (Title: On the total number of deranged partitions).

June 2021 — Speaker at the **National Conference on Applied Mathematics and Didactics**, Ecole Normale Supérieure El Katiba Assia Djebar, Constantine, Algeria. (Title: A combinatorial extension of the Stirling numbers).

May 2021 — Speaker at the **1st International Conference on Pure and Applied Mathematics**, University of Kasdi Merbah, Ouargla, Algeria. (Title: The deranged Bell numbers).

February 2019 — Speaker at the international conference **TAMTAM**, University Abu Bekr Belkaid, Telemcen, Algeria. (Title: On central Fubini polynomials).

December 2018 — Attended the first PhD students' day **Advanced technologies for electronic systems, sustainable energies, telecommunications, and signal/image processing**, The National School of Electronics and Telecommunications, Sfax, Tunisia.

September 2018 — Organizer at **Trees Structure in Theoretical Informatics and Enumerative Combinatorics Days**, University of Science and Technology Houari Boumediene, Algiers, Algeria.

April 2018 — Speaker at the **Scientific Days of Laboratory**, University of Science and Technology Houari Boumediene, Algiers, Algeria. (Title: On a new Fubini-type polynomials).

February 2018 — Speaker at the **Seminar of Laboratory**, University of Science and Technology Houari Boumediene, Algiers, Algeria. (Title: Une classe de polynômes de type Fubini).

May 2017 — Speaker at the **Seminar of Laboratory**, University of Science and Technology Houari Boumediene, Algiers, Algeria. (Title: Quelques propriétés arithmétiques des polynômes de Fubini).

April 2017 — Organizer at **SageMath Days**, University of Science and Technology Houari Boumediene, Algiers, Algeria.

Speaker at the **Scientific Days of Laboratory**, University of Science and Technology Houari Boumediene, Algiers, Algeria. (Title: Congruence for geometric polynomials).

Grants and awards

- **PNE grant,** (December 2019-February 2021)
  Awarded through the **Exceptional National Program (2019-2020)** to visit Sopron University, Hungary, for 10 months.
- **Award from the Minister of National Education,** (July 2024)
  Received for being part of the organizing team in the **First Algerian Mathematical Olympiad (ALMO24)**.

Scientific visits

September 2025	Research visit to The Department of Mathematics and Computer Science "U. Dini" (DIMAI) at the University of Florence, Florence, Italy, lasting two weeks.
February 2025	Research visit to Akdeniz University, Antalya, Turkey, lasting two weeks.
February 2024	Research visit to Akdeniz University, Antalya, Turkey, lasting two weeks.
February 2023	Research visit to Sopron University, Sopron, Hungary, lasting two weeks.
Dec 2019-Feb 2021	Extended research visit to Sopron University, Sopron, Hungary, spanning 14 months.
Jun-Jul 2019	Research visit to Sfax University, Tunisia, Sfax, lasting one month.
Nov-Dec 2018	Research visit to Sfax University, Tunisia, Sfax, lasting one month.

Research Interests

- **Enumerative Combinatorics**
  Set Partitions, Special Sequences, Patterns Enumeration, Permutations, …
- **Analytic Combinatorics**
  Symbolic methods, Generating functions, Recurrences, Asymptotics, …
- **Graph Theory**
  Algorithms, Graph Coloring problems, Graph Sequences, …

Skills

Languages	Arabic: mother tongue. English: very good. French: very good. German: basics.
Coding	Python, C & C++, MATLAB, Mathematica, SageMath, LaTeX.
Web Dev	HTML, CSS, JavaScript.

References

- **Prof. Hacène Belbachir**
  Professor
  University of science and technology Houari Boumediene, BP 32, El Alia, Bab Ezzouar, 16111 Algiers, Algeria.
  hacenebelbachir@gmail.com

References (continued)

- **Prof. Ayhan Dil**
  Associate Professor
  Akdeniz University, Department of Mathematics, 07058 Antalya, Turkey.
  adil@akdeniz.edu.tr
- **Dr. Németh László**
  Dr. habil.
  Sopron University, 9400 Sopron, Bajcsy-Zsilinszky street 4.
  Nemeth.laszlo@uni-sopron.hu